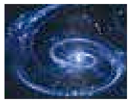
Universal bandwidth boost
Method discovered by Quantum Physicists to transmit double the amount of information using a single particle **http://dgit.in/ubwidth**



A universe in a lab
To better understand the origins of the universe, scientists are simulating conditions seen in the early universe, in a lab **http://dgit.in/labuniverse**

EXOTIC STATES OF MATTER

Researchers are creating strange conditions in laboratories and coming across freakish states of matter all the time. These are the new states of matter identified recently.

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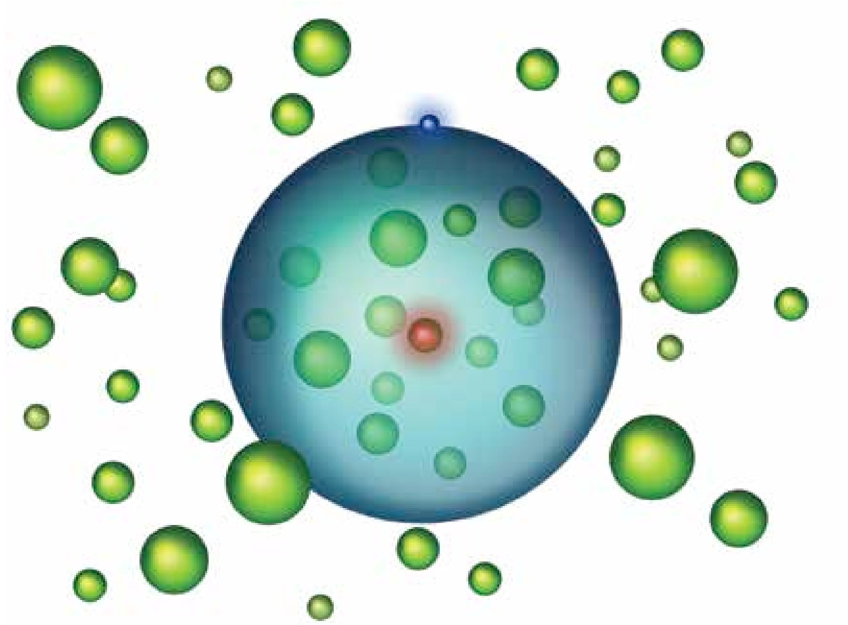

A Rydberg Polaron

A state of matter is a description of the properties of a substance or material. There are four classical states of matter – solid, liquid, gas and plasma. Solids have definite shape and volume, liquids have a definite volume but change their shape according to the container, and gases have no definite shape or volume. A gas will fill all available space in a container. Plasma, like gas, has no fixed shape or volume, and is electrically conductive. It is produced by phenomena such as lightning and some kinds of fires. These are the states of matter that humans are likely to come across on an everyday basis. However, in extreme environments, or under special conditions, exotic states of matter are known to exist. There are over 30 known states of matter, and those listed below are only the newest to be identified.

RYDBERG POLARONS

There is a lot of subatomic space. A hydrogen atom is made up of just one proton and one electron. If your fist is the electron, then the dome of a cathedral or the Gol Gumbaz is the size of the entire atom. The electron is a tiny moth within this vast space. In February 2018, an international team of researchers from Austria and the USA proved that this new state of matter, called “Rydberg polarons” can exist. The process involved freezing strontium atoms close to absolute zero temperatures, creating another state of matter known as a Bose-Einstein condensate. Now, one of these strontium atoms was fired at with a laser, providing it with energy. The single strontium atom expanded to thousand times the size of a hydrogen atom, with the other strontium atoms enclosed within the path of the electron of that atom, known as the Rydberg electron. The other strontium atoms exert a very slight force on the Rydberg electron, barely deflecting it from its path. At higher temperatures, this bond would break, so Rydberg polarons can only be observed at extremely low temperatures.

QUANTUM SPIN LIQUID

Quantum spin liquid is a state of matter that is naturally found on the Earth. In regular magnetic materials, the spins of electrons interact with each other and settle in an orderly fashion. When these magnets are cooled to near zero temperatures, the spins of electrons freeze. In frustrated magnets, the electron spins never settle, even at close to sub zero temperatures. They perpetually interact with each other in a liquid like state, which lends the name to the state - quantum spin liquid. The existence of quantum spin liquids was first proposed in the